WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-08-24 to 2026-08-30

Curated articles in political communication and computational social science.

SUMMARY

COMAP Journal Bot found a week organized around the politics of credibility, technological adoption, and public response. The strongest new work connects partisan sorting to epistemic inequality, traces asymmetric processing of political falsehoods, compares campaign issue strategies across Facebook, Instagram, and TikTok, and tests how project design can bridge ideological opposition to renewable energy.

The broader set adds useful work on public attitudes toward embryo selection and AI infrastructure, computational policy modeling, institutional learning, generative-media attribution, and governance under climate and technology uncertainty.

- New This Week: 17 papers spanning political communication, misinformation, public opinion, governance, and computational policy analysis.
- Catch-Up: 21 papers, including the prior week's reviewed set plus newly recovered work on generative AI attribution, machine-learning audits of aid inequality, democracy, and climate policy.
- Late Additions: 6 papers on smartphone alerts, data-center politics, carbon accounting, urban heat, telecommunications resilience, and biodiversity governance.
- Timing note: This digest was prepared on Saturday, 2026-08-29, before the nominal window closed on Sunday, 2026-08-30; later publications may be recovered in a subsequent catch-up or late-additions section.

HIGHLIGHTS

Partisan bias in a sorted party system: A problem of epistemic inequality

AMERICAN JOURNAL OF POLITICAL SCIENCE · 2026-08-26

Why it matters · Moves the explanation of partisan distrust beyond individual bias to show how party sorting creates credibility deficits and testimonial injustice, especially for marginalized groups.

Open article →

Less Effort to Believe and More Effort to Disbelieve: Examining the Interplay between News-Finds-Me and Affective Polarization in Political Fake News Susceptibility in the United States

COMMUNICATION RESEARCH · 2026-08-26

Why it matters · Combines survey and response-latency evidence to show how News-Finds-Me beliefs and affective polarization produce asymmetric effort when people accept or reject political falsehoods.

Open article →

Social Media Savvy? The Use and Effectiveness of Political Parties' Issue Strategies Across Platforms in the German Federal Election Campaign 2025

POLITICAL COMMUNICATION · 2026-08-26

Why it matters · Links manual content analysis with election-survey data across Facebook, Instagram, and TikTok, showing that engagement rises when party issue strategies fit platform-specific audiences.

Open article →

From Exposure to Creation: Cultivating Narrative Ownership to Reduce Stigma

COMMUNICATION RESEARCH · 2026-08-24

Why it matters · Extends narrative-persuasion theory from exposure to creation, identifying psychological ownership as a mechanism through which writing stories can reduce stigma.

Open article →

Agrivoltaics can reduce political polarization and local opposition to solar energy on land

NATURE COMMUNICATIONS · 2026-08-27

Why it matters · A preregistered field experiment and conjoint study with 2,145 Swiss residents shows how local ownership and project design can reduce ideological polarization over solar development.

Open article →

Public perceptions of polygenic testing and embryo selection for non-medical traits

NATURE HUMAN BEHAVIOUR · 2026-08-26

Why it matters · A cross-national registered report compares stated attitudes with forced-choice decisions, revealing how people actually use medical and non-medical polygenic information in embryo selection.

Open article →

Multi-source agent-based modeling to optimize influenza mitigation strategies in Hong Kong

NATURE COMMUNICATIONS · 2026-08-25

Why it matters · Demonstrates a multi-source agent-based modeling workflow that integrates surveillance, serology, and school absenteeism data to evaluate signal-responsive public-health interventions.

Open article →

How sovereign control, decarbonization and energy costs shape public support for data centers

NATURE COMMUNICATIONS · 2026-08-10

Why it matters · A conjoint experiment shows that public support for data centers depends on digital sovereignty, operator nationality, decarbonization, and local economic costs-an important politics-of-AI infrastructure result.

Open article →

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Previous Week Catch-Up

- American Journal of Political Science
- Communication Research
- Nature
- Nature Communications
- Nature Human Behaviour
- Political Communication
- Proceedings of the National Academy of Sciences
- Science
- Science Advances

Late Additions

- Nature Communications

COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
398	167	188
LATE ADDITIONS CANDIDATES	**CURATED DIGEST BELOW**	**NOTE**
43	17 new, 21 catch-up, 6 late additions	The counts above reflect the complete collected set before manual relevance filtering, while the attached PDF and full HTML reflect the curated digest.

FULL CURATED DIGEST

New This Week

American Journal of Political Science

Partisan bias in a sorted party system: A problem of epistemic inequality

2026-08-26 · DOI 10.1111/AJPS.70086

Justin Pottle

Department of Political Science Loyola University Chicago Chicago Illinois USA

Abstract. Abstract Normative and empirical scholars typically treat partisans' distrust of their political rivals as a product of individual cognitive biases. But even if bias is inevitable, it is parties' coalitionbuilding and messaging strategies that determine who bias is directed toward. This paper argues that the social sorting of party conflict exacerbates the epistemic and democratic harm of partisan distrust by generating distinctive patterns of epistemic inequality. I show that when voters associate parties with particular social groups, members of those groups suffer greater risk of testimonial injustice and related credibility deficits owing to their perceived partisanship. These dynamics are most damaging for members of marginalized groups, as party-group overlap can reinforce the prejudices they already face. Addressing these harms, I argue, requires looking beyond individual partisans to the background features of party systems that shape how elites mobilize group identities and animus in the struggle for office.

Read article →

Communication Research

Less Effort to Believe and More Effort to Disbelieve: Examining the Interplay between News-Finds-Me and Affective Polarization in Political Fake News Susceptibility in the United States

2026-08-26 · DOI 10.1177/00936502261472023

Yu Tian; Lars Willnat; Baohua Zhou

Fudan University, Shanghai, P. R. China; Syracuse University, Syracuse, NY, USA

Abstract. Drawing upon an online survey of 719 US adult partisans incorporating response latency data, we connected News-Finds-Me (NFM) to the partisan-motivated reasoning literature and theorized it as a consensus-based mode of news exposure that leads to different news-processing pathways, where the vested news-processing effort depends on whether the political falsehood aligns with one's partisan identity and surrounding majority opinion. Individual-level analysis and headline-level modeling jointly illuminated the asymmetric interplay between NFM and affective polarization. In partisan-consistent scenarios, high-NFM users devoted limited processing time, falling short of cognitive effort to enable affect-laden, directional believing of partisan-consistent fake news. However, when challenged by partisan inconsistency, high-NFM users dedicated significantly more processing time, which exacerbated the relationship between affective polarization and refutation of partisan-inconsistent falsehoods. Contrarily, such an effort discrepancy, widely documented in the partisan-motivated reasoning literature, was weakened or reversed for low-NFM individuals. Extending the "Effort-Reinforces-Directional-Reasoning" thesis to fake news vulnerability, our findings underscore the paradoxical interplay between NFM and affective polarization: while NFM and affective polarization deviate people from active news-seeking and accuracy-oriented reasoning, their joint dynamics form a type of group thinking that reduces vulnerability to partisan falsehoods-even those aligned with one's partisan attachments.

Read article →

From Exposure to Creation: Cultivating Narrative Ownership to Reduce Stigma

2026-08-24 · DOI 10.1177/00936502261478972

Xun Zhu

University of Kentucky, Lexington, USA

Abstract. Stories are powerful agents of attitude change, including attitudes toward stigmatized groups. However, existing theories have focused on narrative exposure, with limited attention to narrative creation. Creating a story involves simulating characters and events and expressing them through language, which may engage unique psychological processes. This research investigates how and why creating stories shapes stigma. In three experiments, participants wrote, imagined, or read a positive story about a stigmatized character. Study 1 showed that narrative creation, especially writing, increased psychological ownership, which predicted greater transportation and lower stigma. Study 2 replicated these findings using a different stigmatized group. Study 3 showed that psychological ownership continued to explain the effects of narrative creation above and beyond perceived agency. These findings extend theories of narrative persuasion by considering the effects of story creation on intergroup attitudes and introducing psychological ownership as a novel mechanism linking creation to narrative processing and outcomes.

Read article →

Nature

Will Meta's deal make social media safe for kids? Researchers are dubious

2026-08-28 · DOI 10.1038/D41586-026-02733-9

Kaia Glickman

Abstract. Abstract unavailable.

Read article →

Are pro-nuclear politics a threat to public safety?

2026-08-26 · DOI 10.1038/D41586-026-02509-1

Peter Fairley

Abstract. Abstract unavailable.

Read article →

Nature Communications

Agrivoltaics can reduce political polarization and local opposition to solar energy on land

2026-08-27 · DOI 10.1038/S41467-026-77141-8

Lukas Fesenfeld; Leon Sistek; Simon Montfort; Dionis Anderegg; Jurg Rohrer; Tobias Schmidt

Abstract. Abstract As solar energy deployment on land intensifies, political polarization and local opposition increasingly challenge its social acceptance. We propose that agrivoltaics - a dual land-use technology that combines solar power generation with agricultural production - can help bridge ideological divides and reduce local opposition by minimizing land-use conflicts and offering additional benefits to rural communities. To test this hypothesis, we conducted a pre-registered, field-embedded experiment with a representative sample (2145 Swiss residents) just prior to a national renewable energy referendum, combining a personalized geolocated information treatment on local agrivoltaics potential with a conjoint experiment varying key technology and project design attributes. Here we show that agrivoltaics garners broad

public support across ideological groups and reduces local opposition, especially when projects are well integrated into agricultural landscapes, do not significantly reduce agricultural yields, and are locally owned. These findings underscore the importance of technology and project design in addressing local and right-leaning opposition to land-based solar energy.

Read article →

Multi-source agent-based modeling to optimize influenza mitigation strategies in Hong Kong

2026-08-25 · DOI 10.1038/S41467-026-74857-5

Liping Peng; Yiyang Guo; Nicole N. Y. Tsang; Xiaotong Huang; Hau Chi So; Benjamin J. Cowling; Dennis K. M. Ip; Tim K. Tsang

Abstract. Seasonal influenza control faces challenges from variable vaccine effectiveness and uncertain non-pharmaceutical intervention (NPI) performance. While vaccination remains the primary strategy, its effectiveness varies between vaccine-matched and mismatched seasons. Increased post-COVID-19 NPI acceptance enhances influenza control feasibility, yet optimal combination approaches remain poorly understood. We use an agent-based model to analyze influenza transmission across six seasons in Hong Kong (2009-2013) with varying epidemic characteristics. We integrate surveillance, serological, and school absenteeism data for calibration, enabling accurate estimation of reported and unreported infections. We evaluate age-targeted vaccination, staying home when sick, mask use, and school-based interventions across diverse real-world scenarios. Compared to baseline, child vaccination consistently outperforms other strategies, with targeting those under 12 yielding the greatest population-level attack rate reduction (up to 8.5% relative reduction per 100,000 vaccinated). Among NPIs, 40% mask coverage reduces attack rates by 18%-43%, comparable to 25% of symptomatic individuals staying home. During vaccine-mismatched seasons, combining high-coverage mask use and staying home reduces attack rates by 79%-84%. High-coverage school-based vaccination is more effective than closures, reducing student attack rates by up to 85% versus 33% for 14-day closures. Our multi-source calibration approach provides robust evidence for prioritizing child vaccination and strategic NPI combinations. Modeling six influenza seasons to evaluate mitigation strategies, this study proposes a signal-responsive policy framework. Findings show child vaccination and mask use curb transmission; their combination is vital during vaccine mismatches.

Read article →

Nature Human Behaviour

Public perceptions of polygenic testing and embryo selection for non-medical traits

2026-08-26 · DOI 10.1038/S41562-026-02562-W

Edmond Awad; Clara Colombatto; Joanna Demaree-Cotton; Brian D. Earp; Jim A. C. Everett; Peng Liu; G. Owen Schaefer; Ilina Singh; Dominic Wilkinson; Julian Savulescu

Abstract. Abstract Some advances in reproductive technologies raise substantial ethical and psychological challenges, for example, the use of preimplantation genetic testing to select embryos based on non-medical traits. While studies have explored public willingness to use preimplantation genetic testing for medical or non-medical attributes, stated willingness may not reflect implantation decisions when such information is available. This large cross-national study examined public views on polygenic embryo testing. In a US sample (N = 1,467), participants were more willing to test for medical conditions (for example, heart disease) than non-medical traits (for example, antisocial behaviour or low intelligence), although over half supported testing for non-medical traits. In a forced-choice implantation task, participants used medical and non-medical information to a similar extent when making decisions. Similar patterns were observed in Chinese participants (N = 623). These findings suggest variability in choices across specific traits and conditions, rather than a uniform distinction between medical conditions and non-medical traits. The Stage 1 protocol for this Registered Report was accepted in principle on 7 October 2025. The protocol, as accepted by the journal, can be found at https://osf.io/vb9c2 .

Read article →

Mentalizing and heuristics as distinct cognitive strategies in human teaching

2026-08-26 · DOI 10.1038/S41562-026-02540-2

Sevan K. Harootonian; Thomas L. Griffiths; Yael Niv; Mark K. Ho

Abstract. Teaching is a foundational social behaviour that can result from mentally effortful reasoning or cognitively frugal heuristics. When do people use these strategies to teach? Here we investigated this using behavioural experiments and computational modelling with adult participants recruited via Prolific. Experiment 1 (N = 100) revealed robust individual differences: some participants taught by reasoning about a learner's knowledge, consistent with an optimal Bayesian pedagogy model, while others relied on simple heuristics that do not require mentalizing. In two preregistered follow-up experiments, we found that people persist in using heuristics even when they are no longer effective (experiment 2, N = 253, P < 0.001, rank-biserial r = 0.287, 95% confidence interval 0.149-0.419) but that this tendency is pre-empted when inference about a learner's knowledge is scaffolded using an auxiliary task (experiment 3, N = 759, P < 0.001, partial $${\eta }_{{\rm{p}}}^{2}=0.107$$, 95% confidence interval 0.068-0.148). These results demonstrate sophisticated arbitration between planning and heuristics during teaching and elucidate the more general mechanisms involved in adapting mental effort during social interactions. Harootonian et al. find that people switch between effortful, tailored teaching and easier, simple shortcuts, favouring shortcuts even when unhelpful. Prompting participants to reflect on learners' limitations restores effective, personalized instruction.

Read article →

Refining the role of behavioural and social sciences in pandemic response post-COVID-19

2026-08-25 · DOI 10.1038/S41562-026-02506-4

Maja Graso; Karl Aquino; Stefan Baral; Nathan Berg; Cory J. Clark; Lucian Gideon Conway; Paul Dolan; Steven Grover; Kyriaki Fousiani; Claudia C. Kitz; Dario Krpan

Abstract. Abstract unavailable.

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Political Communication

Social Media Savvy? The Use and Effectiveness of Political Parties' Issue Strategies Across Platforms in the German Federal Election Campaign 2025

2026-08-26 · DOI 10.1080/10584609.2026.2715702

Christoph Ivanusch; Simon M. Luebke; Anna-Katharina Wurst; Jorg Haler

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Abstract. How does the ever-growing relevance of social media shape political campaigning? A vibrant public and academic debate addresses the role of social media in elections and democratic processes. Yet, despite a surge in research, we still know little about the extent to which political parties adjust established issue strategies to the priorities of different social media platforms' audiences and how effective these are in generating user engagement. Drawing on theories of issue competition - issue ownership, riding-the-wave, and issue yield - we analyze parties' social media communication during the 2025 German federal election campaign on Facebook, Instagram, and TikTok. Combining manual quantitative content analysis of party and candidate posts with large-scale survey data from the German Longitudinal Election Study (GLES), we assess both strategic use and effectiveness of issue strategies. Our results show that parties strategically prioritize owned issues and those salient on the general public agenda, often combining both dimensions in issue yield strategies. However, these strategies only partly align with what the specific platform audiences value and reward. User engagement is highest for posts that match platform-specific issue preferences, especially when the issues are salient both to a party's supporters on that platform and to its wider user base. The findings offer new insights into the dynamics of contemporary digital

campaigning and cross-platform party competition.

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Proceedings of the National Academy of Sciences

Perfectibility in science and democracy: Institutions that learn

2026-08-26 · DOI 10.1073/PNAS.2626408123

Harvey V. Fineberg

Department of Health Policy and Management, Emeritus, Harvard T.H. Chan School of Public Health

Abstract. Abstract unavailable.

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Interactive representations of power and status signals in learning social hierarchies

2026-08-25 · DOI 10.1073/PNAS.2532908123

Chunliang Feng; Feilong Liu; Binjie Yang; Edmund Derrington; Yuejia Luo; Chen Qu; Jean-Claude Dreher

Center for Studies of Psychological Application, South China Normal University; Faculty of Health and Wellness, City University of Macau; Guangdong Key Laboratory of Mental Health and Cognitive Science, South China Normal University; Institute for Neuropsychological Rehabilitation, University of Health and Rehabilitation Sciences; Key Laboratory of Brain, Cognition and Education Sciences (South China Normal University), Ministry of Education; Neuroeconomics, Reward and Decision-Making Team, Institut des Sciences Cognitives Marc Jeannerod, CNRS; Philosophy and Social Science Laboratory of Reading and Development in Children and Adolescents (South China Normal University), Ministry of Education; School of Psychology, Shenzhen University; School of Psychology, South China Normal University; The State Key Lab of Cognitive and Learning, Faculty of Psychology, Beijing Normal University; University Claude Bernard Lyon

Abstract. Social hierarchies, which organize our social relationships, have heterogeneous effects on choice behavior. Two central dimensions of social hierarchies, status, which is earned through recognition of competence, and power, which is the capacity to control the outcomes of others, may contribute to different behavioral profiles. Yet, the neurocomputational processes by which status- and power-related signals are learned based on social feedback, and influence subsequent decision-making, remain unclear. Here, we combine functional neuroimaging with computational modeling to examine how individuals acquire knowledge about others' status and power, and how these learned hierarchy signals shape social influence on advice-taking during a risky choice task. We found that the medial prefrontal cortex covaried with status estimates, whereas the amygdala covaried with power estimates. Moreover, estimates in one dimension influenced choices in the other during learning. This indicates interactions between status and power signals, with retrosplenial cortex activity associated with hierarchy signals beyond immediate choice demands. Finally, status broadly modulated response biases toward both safe and risky advice, whereas power primarily increased bias toward safe advice. These findings characterize how experimentally instantiated status and power signals are learned and influence risky decision-making, indicating partially separable yet interactive representations of these hierarchy signals.

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Science

The worst energy policy in the world

2026-08-27 · DOI 10.1126/SCIENCE.AEJ2018

Paasha Mahdavi; Michael L. Ross

Michael L. Ross is a professor in the Department of Political Science and the Institute of the Environment and Sustainability at the University of California, Los Angeles, Los Angeles, CA, USA.; Paasha Mahdavi is an associate professor in the Department of Political Science and (by courtesy) the Bren School of Environmental Science and Management at the University of California, Santa Barbara, Santa Barbara,

CA, USA.

Abstract. The Iran war could serve as a turning point in accelerating the global transition away from fossil fuels: Every additional day that oil and gas prices are high and future supplies are uncertain, wind and solar power-cheaper and more scalable than ever before-grow all the more appealing. But nearly half of the world's countries have adopted emergency subsidies that research shows will make this transition slower, more expensive, and less equitable.

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SciLine helps journalists bring scientific evidence to midterm election news coverage

2026-08-27 · DOI 10.1126/SCIENCE.AEL7924

Carmen Nobel

Carmen Nobel, previously the director and editor-in-chief of The Journalist's Resource program at Harvard's Shorenstein Center, is helping lead SciLine's election coverage project with SciLine's journalism projects editor, Elena Renken.

Abstract. Abstract unavailable.

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Correcting the logic behind the 2026 US vehicle emissions standard rollback

2026-08-27 · DOI 10.1126/SCIENCE.AEF0464

Antonio M. Bento; Kenneth T. Gillingham; Mark R. Jacobsen; Christopher R. Knittel; Benjamin Leard; Joshua Linn; David S. Rapson; James M. Sallee; Arthur A. van Benthem; Kate S. Whitefoot

Department of Agricultural and Resource Economics, University of Maryland, College Park, MD, USA.; Department of Economics, University of California, Davis, Davis, CA, USA.; Department of Economics, University of California, San Diego, La Jolla, CA, USA.; Department of Economics, University of Tennessee, Knoxville, TN, USA.; Department of Engineering and Public Policy, Carnegie Mellon University, Pittsburgh, PA, USA.; Haas School of Business, University of California, Berkeley, Berkeley, CA, USA.; National Bureau of Economic Research, Cambridge, MA, USA.; Resources for the Future, Washington, DC, USA.; Sloan School of Management, Massachusetts Institute of Technology, Cambridge, MA, USA.; The Wharton School, University of Pennsylvania, Philadelphia, PA, USA.; USC Price School of Public Policy, University of Southern California, Los Angeles, CA, USA.; Yale School of the Environment, Yale University, New Haven, CT, USA.

Abstract. Correction of selective assumptions and inconsistent interpretations of evidence would overturn policy justification

Read article →

Science Advances

Identifying heat-related diagnoses in emergency department visits among adults in Chicago: A heat-wide association study

2026-08-28 · DOI 10.1126/SCIADV.AED3977

Hyojung Jang; Peter M. Graffy; Benjamin W. Barrett; Daniel E. Horton; Jennifer L. Chan; Abel N. Kho

Defusing Disaster Working Group, Buffett Institute for Global Affairs, Northwestern University, Evanston, IL 60208, USA.; Department of Earth, Environmental, and Planetary Sciences, Northwestern University, Evanston, IL 60208, USA.; Department of Emergency Medicine, Feinberg School of Medicine, Northwestern University, Chicago, IL 60611, USA.; Department of Medicine, Feinberg School of Medicine, Northwestern University, Chicago, IL 60611, USA.; Division of Biostatistics & Informatics, Department of Preventive Medicine, Feinberg School of Medicine, Northwestern University, Chicago, IL 60611, USA.

Abstract. Extreme heat is an escalating public health concern. Although prior studies have examined heat-health associations, their reliance on restricted diagnoses and diagnostic categories may miss or misclassify heat-related illness. Here, we conducted a heat-wide association study to evaluate acute-care diagnoses associated with extreme heat in Chicago, Illinois. Using 916,904 acute-care visits-including emergency department and urgent care encounters-among 372,140 adults across four health care systems from 2011 to 2023, we applied a two-stage analytic approach: quasi-Poisson

regression to screen diagnosis codes associated with heat, followed by distributed lag nonlinear models in a time-stratified case-crossover design to characterize exposure-response and lag patterns. Stage 1 retained 44 diagnosis codes, and 33 clinically reportable diagnoses were retained after stage 2. We observed elevated cumulative odds over lags 0 to 3 for multiple diagnoses, including fluid-balance and renal disorders, dermatologic and pressure-related conditions, multiple sclerosis, a cannabinoid-related mental and behavioral disorder, varicose veins of the lower extremities, and multiple injuries and external causes. By analyzing the full diagnostic spectrum of acute-care services, this study improves the clinical characterization of heat-associated morbidity in acute-care settings.

Read article →

Previous Week Catch-Up

American Journal of Political Science

A theory of political divergence and gradual institutional change

2026-08-20 · DOI 10.1111/AJPS.70084

Clair Yang

The Henry M. Jackson School of International Studies, College of Arts and Sciences, University of Washington Seattle Washington USA

Abstract. Abstract Why do some societies succeed in breaking away from historical traditions while others remain trapped? This paper proposes a formal theory to investigate the interaction between shortrun and longrun political dynamics, highlighting a coevolution of political institutions and social structures. A society consists of various social groups, such as the rulers, the elites, and the commoners. Shortrun strategic considerations give rise to different coalitions among these groups, which determine not only political regimes but also redistribution policies. In the long run, these policies shape the accumulation of wealth and power, where small initial differences can lead to divergent paths of development. Finally, the paper examines three historical case studies on medieval Europe, imperial China, and the medieval Muslim world to illustrate potential pathways of longrun political development.

Read article →

Captured participatory institutions: Theory and evidence from Venezuela's Consejos Comunales

2026-08-18 · DOI 10.1111/AJPS.70077

Jared Abbott; Jose MoralesArilla

Department of Political Science California State UniversityLos Angeles Los Angeles California USA; Escuela de Gobierno y Transformacion Publica Tecnologico de Monterrey Ciudad de Mexico Mexico

Abstract. Abstract Can participatory institutions (PIs) facilitate development when captured by the government? We propose an original theoretical framework to investigate how changing incumbent priorities can shape the developmental and political effects of captured PIs. While patrons concerned with survival can force PIs to specialize in political activities, those pursuing regime consolidation may incentivize PIs to multitask, delivering developmental gains and political rewards. We deploy our framework to study Venezuela's Communal Councils. Leveraging differenceindifferences analyses, an original survey of Council participants, and 51 interviews with government officials and grassroots Communal Council members, we find that Councils yielded both economic and political returns during an early period marked by political stability and regime consolidation. However, as Venezuela's economic collapse pushed the government into an authoritarian survival mode, Councils became politically specialized. While showing that captured PIs can foster development under certain circumstances, our findings cast doubt on PIs' developmentenhancing potential under autocracy.

Read article →

Communication Research

What Do We Mean by Transparency? Conceptualizing and Operationalizing Transparency in Communication

2026-08-17 · DOI 10.1177/00936502261470088

Wenjia Yan; Li Crystal Jiang; Yiqian Gao; Pingping Jia

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Abstract. By analyzing 164 articles from communication journals published between 2002 and 2024, this scoping review systematically investigates temporal trends, conceptualizations, and operationalizations of transparency research within communication studies. The analysis identifies three primary conceptual dimensions of transparency: (1) information, encompassing both visibility and interpretation; (2) relations, functioning as an intermediary between dialogue and power; (3) values and norms, reflecting the tensions between ethical ideals and institutional constraints. Operationalization of transparency has predominantly relied on informational measures, whereas relational and value-based dimensions remain comparatively underdeveloped. Increasing attention has been directed toward subjective stakeholder perceptions, emphasizing transparency as a dynamic and context-dependent construct. Overall, the findings indicate that transparency extends beyond a stable informational attribute to incorporate relational and normative functions, while its enactment remains fluid, contextually shaped, and susceptible to unintended consequences.

Read article →

Nature

Asymmetric prefrontal representations for leader-follower dynamics

2026-08-19 · DOI 10.1038/S41586-026-10900-1

Yuan Cheng; Yusi Chen; Myungji Kwak; Ross P. Kempner; Rudramani Singha; Jared Winslow; Runqi Liu; Umais Khan; Tessa Spangler; Alvi Khan; Talmo Pereira; Matthew Whiteway; Evan S. Schaffer; Nuttida Rungratsameetaweemana; Nan Yang; Herbert Zheng Wu

Abstract. Across species, cooperative behaviour is often organized by distinct social roles such as leaders and followers1, yet the neural mechanisms that support these emergent role dynamics remain unclear. Here we introduce a mouse paradigm that captures leader-follower dynamics during cooperation. In this paradigm, stable social roles emerge through reciprocal interaction and predict learning speed. Disrupting the activity of the medial prefrontal cortex (mPFC), particularly in followers, impairs cooperation and induces complementary shifts in how animals weigh self- and partner-related cues during decision-making. Calcium imaging reveals that the mPFC represents leader-follower dynamics and computes an egocentric social value map of the partner's position in a role-dependent manner. By integrating these empirical findings with a multi-agent inverse reinforcement learning framework, we identify latent value functions that guide cooperative decisions and are decodable from mPFC activity. These findings identify prefrontal representations of leader-follower dynamics and partner information, revealing how social roles structure asymmetric yet reciprocal influence over joint decisions.

Read article →

Nature Communications

What is the problem and what can be done when data sharing challenges public legitimacy?

2026-08-21 · DOI 10.1038/S41467-026-77006-0

Klaus Hoeyer; Lea Skovgaard

Abstract. Surveys suggest that health data sharing with industry and across national borders challenges public legitimacy. We argue that this resistance reflects deeper concerns about profit motives and loss of control and provide guidance on responsive governance to ensure public trust.

Read article →

Tracking funding disparities in global health aid with machine learning

2026-08-18 · DOI 10.1038/S41467-026-76542-Z

Finn Sturenburg; Kerstin Forster; Nicolas Banholzer; Malte Toetzke; Kenneth Harttgen; Stefan Feuerriegel

Abstract. Abstract Reducing the global burden of disease is crucial for improving health outcomes. However, misalignment between health aid and country-level disease burden leaves vulnerable populations without the necessary support for major health challenges, particularly in the least developed countries. Here, we develop a machine learning pipeline using large language models to track flows in official development assistance (ODA) earmarked for health and identify aid-burden misalignment. We classified 3.7 million development aid projects from 2000 to 2022 (USD ~ 332 billion) into 17 major categories of communicable, maternal, neonatal, and nutritional diseases (CMNNDs) and non-communicable diseases (NCDs). We compared the rank of per capita ODA disbursement against the rank of disease burden, measured in disability-adjusted life years (DALYs). We interpret DALY-based alignment as a policy-relevant heuristic rather than a prescriptive allocation criterion. Although funding and disease burden are significantly correlated for many diseases, there are notable disparities. For example, NCDs account for 59.5% of global DALYs but received only 2.5% of health-related ODA over the study period. This is concerning because low- and middle-income countries face an increasing double burden from both CMNNDs and NCDs. Our results show aid-burden misalignment across multiple diseases in several regions, including Central Africa and parts of South Asia and West Africa. Overall, our results identify health disparities to potentially inform policy decisions on development assistance and support targeted allocation of health aid.

Read article →

Outputs of generative diffusion models are often unattributable

2026-08-18 · DOI 10.1038/S41467-026-75667-5

Zheng Dai; David K. Gifford

Abstract. Abstract Modern generative diffusion models work by replicating the statistical patterns of large training datasets. Developing a method to attribute generated outputs to influential training data would greatly advance our understanding of and ability to regulate these models, leading to much work towards this goal. But is this possible? Here, we show that models trained with enough data often generate samples that are unattributable. We establish this through a large-scale analysis of what-if scenarios, revealing that we can often omit any sample or creator from the training data without affecting a generated sample. Our study focuses on diffusion models, which has become the dominant model for generating audiovisual media, and is also prevalent in many scientific applications including protein structure modeling and therapeutic discovery. Central to our analysis is a model ablation methodology that allows efficient removal of training examples from a trained model without the need to retrain.

Read article →

Decarbonizing the EU cement industry under technology and policy uncertainty

2026-08-17 · DOI 10.1038/S41467-026-76611-3

Paul Tautorat; Darius Sultani; Natalija Ljubic; Michael Pahle; Bjarne Steffen; Tobias S. Schmidt

Abstract. Abstract Cement production is a major source of CO 2 emissions. In the European Union, the Emissions Trading System is the cornerstone instrument to incentivize industry decarbonization, including cement. However, major uncertainties prevail regarding allowance prices, carbon-capture investment costs, and the availability of CO 2 pipelines and storage sites.

These uncertainties affect real-world investment decisions, but are not yet well represented in models. Here, we develop a real options valuation model to assess how 179 cement plants across Europe respond to these uncertainties. Our results show that uncertainty perception strongly influences the timing and geography of investments. High confidence in favorable investment determinants could lead to widespread deployment of carbon capture and storage and biomass use, turning the cement industry into a carbon sink with 0.6 gigatons CO2 removed between 2025 and 2050 in total. Low confidence, however, could lead many cement producers to delay action, resulting in cumulated emissions of 0.8 gigatons CO2 over the same time frame. Biomass availability, financing conditions, and plant location further shape investment behavior, underscoring the need for credible energy and climate policy.

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Nature Human Behaviour

An agenda for real-world fairness research to inform policy and organizations

2026-08-17 · DOI 10.1038/S41562-026-02557-7

Ernst Fehr; Keyu Wu

Abstract. Abstract unavailable.

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Political Communication

Rethinking Assumptions About Moral Content Selection: Moral-Emotional Content May Be Less Appealing Than Assumed

2026-08-21 · DOI 10.1080/10584609.2026.2715701

Isabella Rebasso; Yanna Krupnikov

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Abstract. Recent work has shown that moral-emotional content spreads rapidly on social media platforms such as X (Twitter). These patterns have led to concerns that social media facilitates the spread of moral outrage - potentially exacerbating political conflict and animosity. While most past work has focused on the supply side of this phenomenon, i.e. the transmission of morally charged political content, we consider the demand side: do people actually want to see this type of content in their information environment? In this paper, we first theoretically contextualize the motivations to engage with or avoid moral-emotional content. Then, through three pre-registered studies conducted in the U.S. in 2022 and 2023, we test whether people actively choose information that includes moral-emotional elements. Contrary to our initial expectations, we find that participants do not actively select moral-emotional content compared to neutral or emotional content. Despite previous findings on transmission, our results suggest that people may not want to see this type of content in their information environment. By focusing on selection rather than transmission, the findings presented in this paper add another nuance to an ongoing debate on the prevalence and consequences of online political outrage.

Read article →

Hold That Thought: Enhancing the Durability of Fact-Checking Effects Through Low-Cost Accuracy Reminders

2026-08-21 · DOI 10.1080/10584609.2026.2716761

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University of California; Yonsei University

Abstract. Prior evidence suggests that fact-checking can update beliefs regarding factually incorrect information in the short term but its effects often decay rapidly. Such decay raises normative concerns about its sustainability and real-world impact. Here, we examine whether a cognitively efficient, low-cost intervention - called an "accuracy reminder" - can enhance the durability of fact-checking effects. In a large-scale pre-registered panel longitudinal experiment (N = 7,188), participants were randomly assigned to receive fact-checks and, under some conditions, follow-up prompts to reevaluate the same claim at interval-optimized time points. Across topical domains and partisan identities, reminder conditions produced durable belief correction effects lasting up to four weeks, more than doubling the persistence observed without reminders. Effects were robust across reminder frequency and timing, as well as against individual differences, suggesting implementation flexibility and theoretical generalizability. These findings identify retrieval-based reinforcement as a scalable means of sustaining factual belief correction and reinforcing shared factual baselines in spite of cognitive decay and partisan resistance.

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Assessing Civic Journalism Training: A Mixed-Method Applied Research Study

2026-08-18 · DOI 10.1080/10584609.2026.2715698

Joshua P. Darr; Susan Robinson

Syracuse University; University of Wisconsin

Abstract. This mixedmethod applied research study examines how audiences are responding to news stories that are shaped by retraining efforts to help American journalists rebuild audiences. We wanted to know: were these trainings working? Do audiences choose stories that were influenced by these trainings over more traditional elite-focused, problem-oriented, conflict-driven stories? Working with three newsroom-training organizations, we report the results of a series of conjoint experiments that test how well stories abiding by new journalistic techniques of engagement, solutions and "good-conflict" framing play with audiences. We then use the findings from seven focus groups (n = 49) with surveys to explain why people may have responded the way they did in the tests. We found that audiences overwhelmingly selected and appreciated the re-framed content and reported that it makes them trust the news brands more. The focus group participants claimed such content would entice them to become members, subscribe, or even donate to the mainstream brand - if they came across it. However, only a few of the participants said they had heard about the news outlet. Though the principles behind these trainings may seem counterintuitive to some journalists, they offer a potential path forward in uncertain times for local news.

Read article →

Proceedings of the National Academy of Sciences

Urban tree canopy fragmentation and child mortality in low- and middle-income countries

2026-08-21 · DOI 10.1073/PNAS.2606669123

Dengkai Chi; Daniel Richards; Gabriele Manoli; Jun Yang; John S. Ji; Ye Zhang; Brenda Lin; Amy Hahs; Congqiang Liu; Yue Zhu; Yeshan Qiu; Jing Wang; Paolo Burlando; Simone Fatichi; Puay Yok Tan

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Abstract. Exposure to urban green spaces is associated with lower risks of several leading causes of under-5 mortality (U5M). While urban development improves infrastructure, healthcare access, and socioeconomic conditions, rapid urbanization in low- and middle-income countries (LMICs) has also accelerated the loss and fragmentation of green spaces.

Such fragmentation may weaken ecosystem services critical for child health and increase exposure to vector-borne infections, yet its implications for U5M remain poorly understood and must be weighed against the benefits of urbanization. Here, we quantified associations between urban tree canopy cover and canopy aggregation with all-cause U5M using individual-level longitudinal mortality data between 2000 and 2019 from 57 LMICs, and identified priority areas for expanding canopy cover and improving canopy aggregation across 98 LMICs using counterfactual scenario analyses. The association between canopy cover and U5M was nonlinear, with risk declining steeply at low-to-intermediate levels of canopy cover. At a given level of canopy cover, every 1% increase in tree-to-tree adjacency was associated with a 0.17% reduction in mortality risk (hazard ratio: 0.9983, 95% CI: 0.9972 to 0.9995). The largest contiguous high-priority areas were concentrated in the West Sudanian Savanna, the eastern Punjab region of Pakistan, and the Ganges-Brahmaputra Delta in Bangladesh, where high U5M burdens align with substantial potential to improve tree canopy. This study establishes a robust association between tree canopy characteristics and U5M across LMICs. We identify canopy cover and fragmentation as potentially modifiable risk factors, highlighting important implications of urban greening design for delivering cobenefits for child health and climate mitigation.

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Advances in small area population estimation in the absence of national census data

2026-08-21 · DOI 10.1073/PNAS.2413993123

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Abstract. Population data at small area scales are essential for effective decision-making, influencing public health, disaster response, and resource allocation, among others. While national censuses remain the cornerstone of population data, they are often constrained by high costs, infrequent collection cycles, and coverage gaps, which can hinder timely data availability. To address these challenges, geospatial statistical approaches using limited microcensus surveys have been demonstrated as a reliable source, but the field has advanced substantially in recent years, with significant developments in both data sources and modeling methodologies. New approaches now leverage routine health intervention campaign data, satellite-derived settlement maps, and bespoke modeling approaches to produce reliable small area population estimates where enumeration is difficult or outdated. Various countries are applying these techniques to support census operations, health program planning, and humanitarian response. This manuscript reviews recent advances in "bottom-up" population mapping approaches, highlighting innovations in input data, modeling methods, and validation techniques. We examine ongoing challenges, including partial observation of buildings under forest canopy, population displacement, and institutional uptake. Finally, we discuss emerging opportunities to enhance these approaches through better integration with traditional data ecosystems, capacity strengthening, and coproduction with national institutions.

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Characterizing the impact of incorporating spatially aggregated human mobility data into infectious disease models

2026-08-18 · DOI 10.1073/PNAS.2533465123

Ronan Corgel; Kyra H. Grantz; Lauren Gardner; Derek A. T. Cummings; Harendra de Silva; Thilini Somaratne; Dhammika Silva; LakKumar Fernando; Amy Wesolowski

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Abstract. Models of infectious disease dynamics should align the spatial scale of mobility data to the scale of travel relevant to inferring disease introduction events and subsequent local transmission. Despite this, the biases of spatially aggregating mobility data on model inferences are rarely explored. Here, we examine the sensitivity of infectious disease modeling results to different spatial scales of human mobility by integrating multiscale mobility data from Sri Lanka into Susceptible, Exposed, Infected, Recovered (SEIR) metapopulation models. Aggregated mobility data were obtained from mobile phone records at three increasingly coarser spatial scales to simulate epidemic spread of an emerging acute respiratory infection. We found that travel was not evenly distributed among nested spatial scales when data were disaggregated, with subunits close to borders exhibiting higher levels of travel relative to the unit. In simulations of disease transmission, these different scales of mobility aggregation had potential to yield wide variations in the estimated spatial invasion timing but not in final epidemic size. Modeled differences in spatial invasion time depended on disease transmission intensity and exogenous factors like the initial seeding location. Our results carry implications for infectious disease modeling best practices and public health response, particularly the policy decisions made from model inferences that were or were not informed by the relevant spatial scale of mobility data such as intervention timing, risk communication, and resource allocation.

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Value misalignment in X's feed algorithm is a reflection of value tensions in engagement

2026-08-17 · DOI 10.1073/PNAS.2610388123

Ziv Epstein; Farnaz Jahanbakhsh; Tiziano Piccardi; Axel Peytavin; Isabel Gallegos; Shardul Sapkota; Dora Zhao; Johan Ugander; Michael S. Bernstein

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Abstract. Social media feed algorithms rank content that is purported to be preferred by users, but the engagement behaviors that drive these algorithms are (at best) indirect proxies for users' explicitly self-stated values. Are the resulting feeds value aligned, and if not, why? We investigate this question by annotating the basic human values expressed in participants' X (Twitter) feeds (N = 715 US users), analyzing the relationship between the posts' value expressions and the posts' amplification in the ranked "For You" Page feed, and then comparing the amplified values to users' own values. We observe that the inventory of posts from followed accounts reflects users' self-stated values-but that there is an overall negative correlation (misalignment) between users' explicit values and the value expressions the algorithm is more likely to amplify. We turn to engagement behavior to understand this misalignment and observe that users' engagement behaviors can be misaligned with their stated values-likely causing the algorithm to learn and reflect these misaligned values. We also detect partisan differences consistent with this theory: While the algorithm amplifies values negatively correlated with both Democrats' and Republicans' self-stated values, they are more misaligned for Democrats. And in fact replying, a heavily weighted form of engagement, is associated with values that are less aligned for both Democrats' and Republicans' self-stated values, and is even more misaligned for Democrats. Taken together, these findings offer a glimpse into the tensions between the values that people hold and those that provoke reactions, and how these value tensions can produce misaligned outcomes.

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Science

Technology is not a replacement for democracy The Rise and Fall of the Artificial State Jill Lepore Liveright, 2026. 336 pp.

2026-08-20 · DOI 10.1126/SCIENCE.AEJ1046

Carissa Veliz

The reviewer is at the Institute for Ethics in AI, University of Oxford, Oxford, UK, and is the author of Prophecy: Prediction, Power, and the Fight for the Future, from Ancient Oracles to AI (Doubleday, 2026).

Abstract. A government run by machines is not inevitable-we have been here and rejected it before

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China's solar expansion policy reduces bird diversity

2026-08-20 · DOI 10.1126/SCIENCE.AEE0747

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Abstract. Could the global transition to renewable energy create a green dilemma that pits carbon reduction against biodiversity conservation? This study examined the effect of policies promoting solar photovoltaics on local avian biodiversity using a panel dataset covering 2344 counties in China from 2014 to 2023. Policies that favored photovoltaic expansion led to reductions in bird diversity, disproportionately affecting wealthier and nondesert regions, as well as widespread species. The mechanism operated primarily through land conversion: Cropland and grassland were transformed into developed areas, reducing the diversity of vegetation. Paradoxically, the leaf area index increased, a pattern we term "inferior greening," whereby diverse natural landscapes were replaced by dense but ecologically homogeneous vegetation. We argue that future photovoltaic development should be accompanied by strict biodiversity safeguards, especially in economically developed regions with high habitat complexity.

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Beyond carbon in renewable energy policy

2026-08-20 · DOI 10.1126/SCIENCE.AEK1460

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Abstract. China's policy-driven solar expansion shows why climate policy appraisal should look at biodiversity impacts

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A Brussels Effect for battery sustainability

2026-08-20 · DOI 10.1126/SCIENCE.AED8133

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Abstract. An EU battery regulation can help drive toward a widely shared evidentiary system for sustainability governance

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Science Advances

Tree selection in urban greening shapes air quality for global cities

2026-08-21 · DOI 10.1126/SCIADV.AEE5583

Xianjun He; Bin Yuan; Yibo Huangfu; Joost de Gouw; Sihang Wang; Suxia Yang; Yuwen Peng; David D. Parrish; Xiaobing Li; Ming Chang; Zhanyou Mo; Kyung-Eun Min; Woohui Nam; Yubin Chen; Xiaoxiao Zhang; Jipeng Qi; Qianqian Xie; Weihua Chen; Alex Guenther; Xuemei Wang; Douglas Worsnop; Thomas Karl; Min Shao

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Abstract. Mitigating ozone pollution remains a pressing challenge for megacities worldwide. Biogenic volatile organic compounds (VOCs) are proposed as pivotal precursors to urban ozone formation. However, direct observational evidence is scarce and the extent of this effect across cities is unclear. Here, by combining an unprecedented eddy covariance flux dataset from Beijing with modeling analysis, we find that urban vegetation contributes over half of organic reactivity essential for ozone production, predominantly driving ozone exceedances during hot periods. The unexpectedly high biogenic influences are attributed to the abundance of urban trees with strong emission potentials of VOCs within the city. Model extrapolations reveal that similar or even greater challenges exist for other cities across Asia and Oceania, where urban greening strategies often favor highly emitting trees. Our findings underscore the critical need for selective urban planning in greening programs to counteract a warming climate.

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Late Additions

Nature Communications

Performance and user response of android's smartphone-based alerts in the 2025 Marmara Ereglisi earthquake

2026-08-11 · DOI 10.1038/S41467-026-76413-7

S. Mostafa Mousavi; Patrick Robertson; Richard M. Allen; Alexei Barski; Robert Bosch; Nivetha Thiruverahan; Youngmin Cho; Tajinder Gadh; Steve Malkos; Boone Spooner; Greg Wimpey; Edward Shi; Marc Stogaitis

Abstract. This study presents a comprehensive evaluation of Google's Android Earthquake Alert (AEA) system during the Mw 6.2 Marmara Ereglisi, Turkiye earthquake. AEA detected the event 5.31 seconds after its initiation, alerting over 16 million users. Warning times for weak shaking (MMI III) reached up to 150 seconds, with a median of 56 seconds. While near-source warning windows were shorter, the system achieved 90% true positives and 99% precision overall. The high density of the phone network enabled faster detection than traditional stations, even for this offshore epicenter. Feedback data shows AEA recipients were highly likely to take protective actions, such as "drop, cover, and hold on," or warn others. Timely alerts substantially increased user engagement, perceived usefulness, and future trust. These results highlight how crowd-sourced technology and behavioral insights can effectively enhance seismic resilience on a massive scale. Traditional sensor networks often have gaps that delay critical earthquake alerts. This study finds that a smartphone-based system detected a Mw 6.2 quake in seconds, alerting millions and increasing life-saving responses.

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How sovereign control, decarbonization and energy costs shape public support for data centers

2026-08-10 · DOI 10.1038/S41467-026-76500-9

Jonas Heering; Erik Voeten

Abstract. The rapid expansion of data centers has created new political challenges. Governments want to build more data centers to reap the benefits from the artificial intelligence (AI) economy and achieve their digital sovereignty agendas. However, data centers require enormous amounts of electricity and water, threaten emission-reduction targets, and often raise local electricity prices. As a result, local communities increasingly mobilize against data center construction. We use a vignette and conjoint survey experiment in Germany to evaluate how publics think about the environmental, economic, and (geo)political tradeoffs that data centers entail. We find that directly priming people with digital sovereignty concerns only marginally increases support for building more data centers. Yet, support varies substantially based on characteristics that people do have strong views about: decarbonization, geopolitics, and local environmental and economic impact. Geopolitical concerns are particularly salient: people strongly favor data centers operated by German or European firms over US or Chinese operators. These nationality effects are large even when compared to the effects of electricity price increases and variation in energy sources. Our findings suggest that public opposition to data centers depends on pre-existing political cleavages and that sovereignty concerns loom large. Investigating how publics think about the environmental, economic, and geopolitical tradeoffs that data centers entail, results show that public opposition depends on pre-existing political cleavages as well as concerns about digital sovereignty.

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How carbon accounting rules shape incentives for hydrogen production

2026-08-10 · DOI 10.1038/S41467-026-75473-Z

Gunther Glenk; Philip Holler; Stefan Reichelstein

Abstract. Abstract Governments around the world have recently adopted policy support programs for hydrogen, tying the level of support to the assessed carbon intensity of the hydrogen produced. Here we compare alternative carbon accounting rules for determining the policy support available for hydrogen in terms of the resulting financial and carbon emissions performance of Power-to-Gas systems. We calibrate our model to reference plants eligible for the production tax credit available under the Inflation Reduction Act in the United States. Contrary to frequently articulated views, more stringent accounting rules generally provide investors with sufficient incentives to invest in Power-to-Gas systems. Nonetheless, even more stringent rules can lead to carbon intensity levels close to those for hydrogen produced from natural gas with carbon capture. Less stringent rules generally entail stronger investment incentives due to higher profitability, but also significantly higher emissions as investors procure more carbon-intensive electricity from the general grid.

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A national-scale assessment of the impact of canals on urban temperatures

2026-08-10 · DOI 10.1038/S41467-026-75731-0

M. D. Tomkins; H. McDonald; J. J. Huck; J. Tippett

Abstract. Abstract Urban populations are particularly vulnerable to heat-related mortality and morbidity. Climate-responsive urban design, which can include development of green and bluespaces, has the potential to offset climate change-related impacts. Compared to greenspaces, the effects of bluespaces on urban temperatures have received comparatively little attention, with uncertainty regarding the magnitude of urban cooling. In this study, we develop an energy balance model to estimate the thermal effects of canals and apply this to the canal network of Great Britain and Ireland. Here we show that the modelled canals provide an average annual daytime cooling effect of ~0.8 C (0.7 - 1 C) within the immediate vicinity of the canal network, with more pronounced cooling during spring and early summer (~1.2 - 1.4 C) and during heatwaves (~1.5 - 2.3

C). While thermal effects are not exclusively positive, our results indicate that canals can improve thermal comfort and reduce heat risk, with related benefits for population health and wellbeing.

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Potential of different area-based governance mechanisms for achieving Global Biodiversity Framework goals

2026-08-03 · DOI 10.1038/S41467-026-76225-9

Pablo J. Negret; Victor J. Rincon-Parra; Kendall R. Jones; Vanessa R. Rathbone; Sidney Novoa; Marvin Quispe; Armando Valdes-Velasquez; German Forero-Medina; Judith Schleicher; Tatsuya Amano; Miguel Saravia; Julie Gwendolin Zaehringer

Abstract. Abstract The Kunming-Montreal Global Biodiversity Framework (GBF) includes a target to protect 30% of land by 2030, identifying other effective area-based conservation measures (OECMs) as complementary to Protected Areas (PAs). However, their contribution to conservation outcomes remains uncertain. Here we show the level of human pressure across the global OECM network and assess the impact of different governance mechanisms on deforestation and associated carbon emissions in the Peruvian Amazon from 2005 to 2021. We evaluate Indigenous Lands (ILs), Non-Timber Forest Product Concessions (NTCs), Logging Concessions (LCs), Mining Concessions (MCs), and PAs using statistical matching. Globally, OECMs show higher human pressure than strictly protected PAs, but similar levels to less strictly protected PAs. In the Peruvian Amazon, PAs are most effective at reducing deforestation (49-53% avoided), followed by NTCs and ILs (33% and 20%, respectively). In contrast, LCs and MCs are associated with increased deforestation (13% and 24%). These findings highlight the complementary role of OECMs and the effectiveness of different governance mechanisms in reducing deforestation and carbon emissions, informing efforts to achieve global biodiversity targets.

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Global vulnerability assessment of mobile telecommunications infrastructure to climate hazards using crowdsourced open data

2026-08-03 · DOI 10.1038/S41467-026-76197-W

Edward J. Oughton; Tom Russell; Jeongjin Oh; Sara Ballan; Jim W. Hall

Abstract. Climate change is intensifying coastal flooding, riverine flooding and tropical cyclones, yet risks to telecommunications infrastructure remain poorly quantified despite growing societal dependence on digital connectivity. We present a global, event-based geospatial assessment of mobile cellular infrastructure vulnerability, combining open crowdsourced asset data with global climate hazard models under historical conditions and future scenarios, RCP4.5 and RCP8.5. We analyse approximately 17 million mobile basestations worldwide, estimating exposure and direct damage across extreme-event return periods. Under RCP8.5 in 2050, a 0.1% annual probability tropical cyclone event affects an estimated 3.7 million basestations, causing USD 4.22 billion in direct damage. Under RCP8.5 in 2080, equivalent coastal flooding affects 268,000 basestations, causing USD 6.44 billion in damage. Riverine flooding affects more assets globally, approximately 2.8 million, but damages remain broadly stable at USD 46-47 billion. Telecommunications infrastructure assets should therefore be systematically incorporated into climate risk assessments. Mobile networks face growing risks from climate hazards. The study finds that floods and tropical cyclones could affect millions of base stations worldwide and cause billions of dollars in damage.

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